The endocrine system

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The endocrine system helps control our bodies by secreting (releasing) hormones from glands. Some hormones only act on specific body parts, others cause a more general response. The principal controller of hormone production is the hypothalamus (a part of the brain). The main glands concerned in reproduction are the pituitary, testes and ovaries.

The pituitary is located at the base of the brain and is less than 1 cm in diameter and 0.5 g in mass. As well as releasing hormones that control growth and reproduction it manufactures others, which, in turn, control many other glands.

The *gonads* are where gametes are produced. The male gonads are the testes and the female gonads are the ovaries. Hormones from the pituitary gland and the steroid hormones from ovary or testis control reproductive function in both sexes. In women, fluctuations of these hormones govern each menstrual cycle, stimulating ovulation and preparing the uterus for pregnancy. The hypothalamus causes the pituitary gland to produce gonadotropins – hormones that stimulate gonads.

The sex steroids also bring about the development of secondary sexual characteristics. In males, androgens cause beard growth, deepening of the voice and increase in muscle size. Oestrogen and progesterone cause breast development and the menstrual cycle in females. The male steroids are anabolic (building up) and are among the banned substances that some athletes have used to encourage muscle development and enhance performance.

1 Give reasons why athletic and sports organisations would ban their members form taking anabolic steroids.

When puberty is reached, humans produce follicle stimulating hormone (FSH) and luteinising hormone (LH). FSH causes the formation of sperm in males and stimulates the development of ova in females. LH causes the production of androgens in males and for females stimulates ovulation. FSH starts off the menstrual cycle.

If an egg is not fertilised the cycle is approximately as follows:

- *First 5 days:* menstruation, the lining of the womb breaks down and is shed.
- *Next 7 days:* an egg matures in the follicle.
- *Next 3 days:* ovulation – egg is released into fallopian tube.
- *Next 7 days:* egg dies if not fertilised.
- *Next 6 days*: uterus lining continues to thicken before it starts to break down.

Complete the following table.

Gland	Location	In male or female	Produces
testis	groin	2	3
4	base of brain	5	range of hormones
ovary	6	7	8

9 What is a *gonad*?

10 Represent the menstrual cycle on the timeline below.

The menstrual cycle

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